Velocity and position graphs

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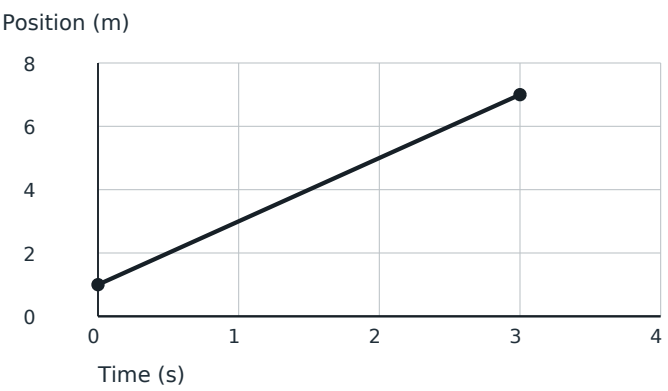
Learning target: Connect a position-time graph to velocity using slope.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 1, Day 2.*
Distance 12 m; displacement +2 m (right); final position 5 m.

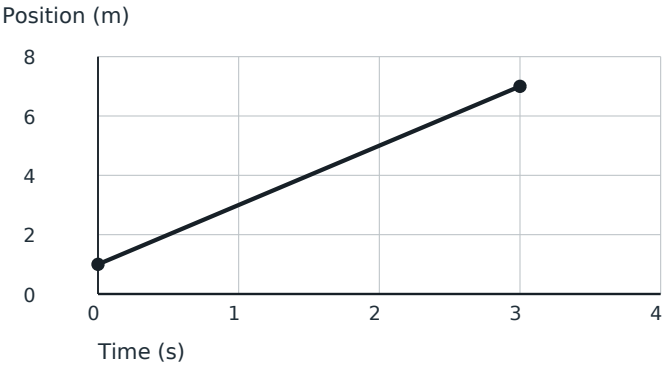
02. Worked example: explain the model
Straight line through (0, 1) and (3, 7), rise 6 m and run 3 s; positive slope.

Model / diagram check



03. Read the motion picture
x(0)=1 m and x(3)=7 m; motion is right/positive.

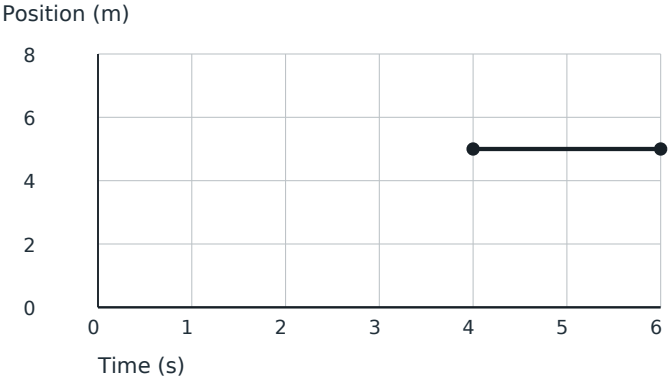
Model / diagram check



04. Find the rate
7-1=6 m; 3 s; +2 m/s/right.

05. A pause
No. Position stays 5 m, so displacement is zero and velocity is zero.

Model / diagram check



06. Draw a walk with a pause
0-2 s: right at 2 m/s; 2-4 s: stopped; 4-6 s: left at 2 m/s. Points (0, 1),(2, 5),(4, 5),(6, 1).

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07. Distance or displacement?
Distance 4+4=8 m; displacement 0 m. Average velocity uses displacement; the walker returns to the start.

08. Review: connect an earlier idea *Review: Unit 1, Day 2.*
Distance 14 m; displacement +2 m (right); final position 6 m.

09. Exit check
v = (3 - 9)/5 = -1.2 m/s, or 1.2 m/s left. The graph is a straight line from (0 s, 9 m) to (5 s, 3 m).

10. STRETCH: test the model
Any two parallel lines with slope 2 m/s; vertical intercepts differ.

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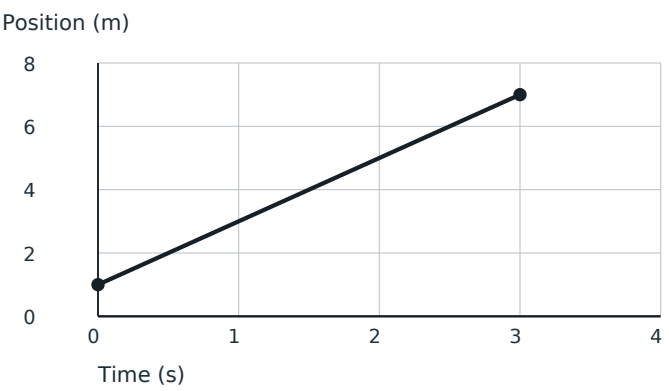
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01. Warm-up: bring back an earlier idea *Review: Unit 1, Day 2.*
Distance 13 m; displacement -3 m; final position +2 m.

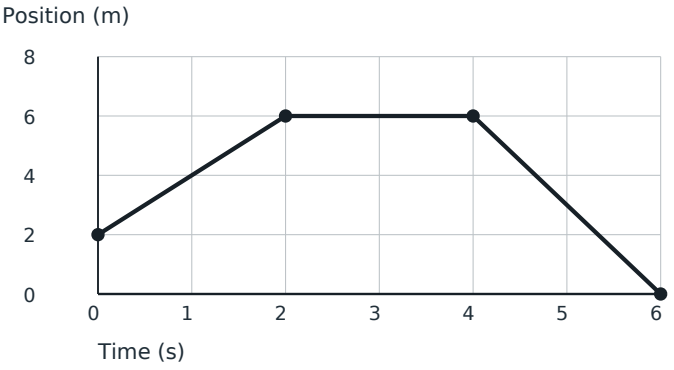
02. Worked example: explain the model
Straight line through (0, 1) and (3, 7), rise 6 m and run 3 s; positive slope.

Model / diagram check



03. Interpret all three intervals
0-2 s: (6-2)/2=+2 m/s; 2-4 s: 0; 4-6 s: (0-6)/2=-3 m/s.

Model / diagram check



04. Compare two averages
Distance 4+6=10 m; average speed 10/6=1.67 m/s. Displacement 0-2=-2 m; average velocity -0.333 m/s.

05. A line that misses the origin
v=-2 m/s; x=18-2 t (m, s); x(9)=0 m, assuming the same line continues.

06. Construct the graph
Vertices (0,-2),(2, 4),(4, 4),(7,-2). Net displacement zero; slope signs match each interval.

07. Separate height and slope
It gives average velocity over 0-5 s if x(0)=0 m. It is also instantaneous velocity at 5 s only if the appropriate local slope is 2 m/s, e.g. a straight line from the origin. A single point is insufficient.

08. Review: connect an earlier idea *Review: Unit 1, Day 2.*
Distance 15 m; displacement +1 m; final position -2 m.

09. Exit check
v=8/4=+2 m/s; position increases by 2 m per second. x(7)=10 m.

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10. STRETCH: test the model
1+2 t=9-t gives t=8/3 s and x=19/3 m. Intersection represents same place and time.